



Roll No:

MCA-INT
(SEM VI) THEORY EXAMINATION 2024-25
GRAPH THEORY

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

 $02 \times 7 = 14$

Q no.	Question
a.	What do you mean by orthogonal vectors?
b.	Define Euler graph.
c.	Calculate the number of edges in the graph G with 16 vertices, each of degree 2.
d.	Prove that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)(n-2)}{2}$ edges.
e.	Find the chromatic number of a complete graph of n vertices.
f.	Define the Mincut Theorem.
g.	What are the applications of a planer graph?

SECTION B

2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question
a.	Write the Kruskal's algorithm for finding the minimum spanning tree of a graph, Illustrate the algorithm using an example.
b.	Prove that a connected planar graph with n vertices and e edges has $n - e + 2$ regions.
c.	Explain thickness, crossing and covering with example. Define five color problems. are there any graphs that cannot be colored with four colors?
d.	Explain geometrical dual and combinational dual graphs.
e.	Describe the steps to find adjacency matrix and incidence matrix for a directed graph with a suitable example.

SECTION C

3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Use Dijkstra's algorithm to compute all shortest paths starting at node s. Show the values of the program variables B, R, U, p, D after each iteration of the main while-loop of Dijkstra's algorithm.



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- b. Prove that a spanning tree T of a given weighted connected graph G is shortest spanning tree of G if and only if there exists no other spanning tree of G at a distance of one from T whose weight is smaller than that of T .

Q no.	Question
a.	When is a graph said to be regular? Show that the number of vertices in a k -regular graph is even if k is odd. Find all non isomorphic simple graphs of order 4.
b.	Explain diameter and radius of a tree with example. Find the condition under which the diameter of a tree is equal to twice the radius.

Q no.	Question	07 x 1 = 07
a.	Discuss how the concept of circuit vectors and cycle spaces is used in applications such as electrical network analysis, coding theory, or topological graph theory. What are the advantages of using such an algebraic approach in analyzing properties of graphs?	
b.	Explain the meaning and significance of the Rank-Nullity Theorem in the context of solving systems of linear equations. Why does this theorem guarantee that the solution space of a homogeneous system $Ax=0$ has a well-defined dimension?	

Q no.	Question	07 x 1 = 07
a.	Explain the significance of the Four Color Theorem in the context of planar graphs. Why is it not trivial to prove, even though it seems intuitive? Discuss the historical challenges in proving this theorem.	
b.	Draw planar connected graph such that i) $e = 3n - 6$ ii) $e < 3n - 6$	

Attempt any <i>one</i> part of the following:		07 x 1 = 07
Q no.	Question	
a.	What are chromatic polynomials? Explain the concept of chromatic partition.	
b.	Find all nonisomorphic simple graphs of order 4.	